

Route 1 — Data Quality Audit & Route Optimization Analysis | FirstKnock Sales OS

Route 1 — Data Quality Audit & Route Optimization Analysis

FirstKnock Sales OS | Operational Review Document

April 2026

Route 1 — 867 Stops | Origin: 911 N Kings Rd, West Hollywood, CA 90069 | Coverage: ~40 sq mi | Generated: April 2026

867Total Stops in File	~62Recommended Removals	~805Clean Stops After Audit	10m Lead — Knock NOW
------------------------	-------------------------	-----------------------------	----------------------

Executive Summary

Route 1 contains significant data quality issues that must be resolved before this route is assigned to any rep. The file contains approximately 52 confirmed duplicate stops across 7 repeated data blocks — entire sequences of consecutive stops appear twice, a known artifact of the 'Combine All' merge function in ZipCodeExplorer. Beyond block duplication, two records are definitively invalid: Stop #490 ('0 N Alta Vista Blvd') carries a house number of zero and is not a real address, and Stop #848 ('911 N Kings Rd') is the route origin itself — not a sales prospect. An additional 8 stops on Hollywood Blvd, the Sunset Strip, and Wilshire Blvd are flagged as likely commercial addresses requiring manual verification before knocking. Total recommended removals are approximately 62 stops, leaving approximately 805 clean, actionable prospects.

Recency distribution is the most operationally damaging issue in this route. Only a single stop carries a '0m' label — Stop #459, 1207 Greenacre Ave, West Hollywood 90046, which sold this month and is the single highest-priority door in the entire route. Approximately 142 stops (16.4%) are fresh (sold within 30 days), and 218 stops (25.1%) are recent (1–3 months). However, 58.4% of stops — approximately 507 — show recency of 4 months or older, which is inconsistent with the default 3-month lookback window. This suggests the route was generated with the 'Sold within' filter set to 12 months rather than 3. Critically, fresh leads are NOT front-loaded in the sequence: they are distributed roughly evenly across all 867 stops (~27% in stops 1–200, ~25% in 201–400, ~23% in 401–600, ~26% in 601–867). This means reps may spend the first half of the day knocking stale 10–12 month leads before reaching fresh prospects — a significant and correctable inefficiency.

Geographic sequencing is highly inefficient. The 2-opt TSP solver was run on the full 867-stop set without neighborhood pre-clustering, which severely degrades TSP performance at this scale. There are approximately 35–45 major teleportation jumps (greater than 2 miles between consecutive stops), and multiple street sweep failures — most critically, Kirkwood Dr is visited at

stops 10–11, abandoned, revisited at stop 35, then again at stop 856, an 845-stop gap. Neighborhood clusters are fragmented: ZIP 90046 appears in 6 non-contiguous clusters, Beverly Hills stops are split across 4 separate clusters, and the route oscillates repeatedly across the same geographic areas. At 867 stops, this mega-route represents 8–34 standard FirstKnock routes worth of data. The recommended remediation is to split into 8 sub-routes by neighborhood cluster, re-pull with a strict 3-month lookback, and re-run routeOptimizer.js independently per sub-route.

PRIORITY ALERT — Stop #459: 1207 Greenacre Ave, West Hollywood, CA 90046. Recency label '0m' — sold THIS MONTH. This is the single hottest lead in the entire 867-stop route. Knock this door before any other stop in the route.

Section 1 — Data Quality Audit

1A. Duplicate Detection

The route file contains repeated data blocks — entire sequences of consecutive stops appear twice in the file. This is not random duplication but block-level repetition, consistent with a UI rendering artifact or a 'Combine All' merge duplication in ZipCodeExplorer where route segments were concatenated without deduplication. The second occurrence of each block should be removed entirely. Approximately 52 stops are affected across 7 confirmed duplicate blocks.

Table 1 — Confirmed Duplicate Data Blocks (approx. 52 stops total)

Block	First Occurrence (Stops)	Addresses Included	Action
Block 1	Stops 137–146	2224 Guthrie Dr, 2261 Bagley Ave, 2400 Guthrie Ct, 2442 S Beverly Dr, 9600 Beverlywood St, 2612 S Beverly Dr, 2623 S Beverly Dr, 2631 Castle Heights Pl, 2721 S Beverly Dr, 2807 Cardiff Ave	Remove second occurrence
Block 2	Stops 172–174	1770 Hauser Blvd, 5518 Venice Blvd, 5522 Venice Blvd	Remove second occurrence
Block 3	Stops 377–386	3001 Dona Nenita Pl through 2155 Hercules Dr (10 stops)	Remove second occurrence
Block 4	Stops 659–668	445 N Lucerne Blvd through 742 N Hudson Ave (10 stops)	Remove second occurrence

Block 5	Stops 694–703	533 S Rimpau Blvd through 607 N Alta Vista Blvd (10 stops)	Remove second occurrence
Block 6	Stops 797–803	324 S Canon Dr through 9674 Wilshire Blvd (7 stops)	Remove second occurrence
Block 7 (Near-Dupe)	Stops 623–624	211 N Western Ave (6m) and 201 N Western Ave (6m) — sequential stops, same block, identical recency	Flag for review

Stop #623 (211 N Western Ave) and Stop #624 (201 N Western Ave) are near-duplicates — sequential addresses on the same block with identical recency labels. These are likely two distinct properties but should be verified before knocking.

1B. Invalid & Suspicious Addresses

The following two stops are definitively invalid and must be removed from the route immediately, before any rep assignment.

- Stop #490: 0 N Alta Vista Blvd, Los Angeles, CA 90036 — House number is 0. This is not a valid residential address. No property exists at this address. Remove immediately.
- Stop #848: 911 N Kings Rd, West Hollywood, CA 90069 — This is the route origin and depot address. It is not a sales prospect. Remove immediately.

The following 8 stops are flagged as likely commercial addresses based on known corridor geography. Manual verification is required before removal — some Hollywood Blvd addresses may be residential units above commercial ground floors.

Table 2 — Flagged Commercial Addresses

Stop #	Address	ZIP	Commercial Corridor	Recommendation
45	8558 Hollywood Blvd	90069	Hollywood Blvd corridor	Manual verify before removal
58	8804 Hollywood Blvd	90069	Hollywood Blvd corridor	Manual verify before removal
59	8800 Hollywood Blvd	90069	Hollywood Blvd corridor	Manual verify before removal
67	8383 Hollywood Blvd	90069	Hollywood Blvd corridor	Manual verify before removal

68	8378 Hollywood Blvd	90069	Hollywood Blvd corridor	Manual verify before removal
803	9674 Wilshire Blvd	90212	Wilshire Blvd (BH commercial)	Remove
858	8800 W Sunset Blvd	90069	Sunset Strip	Remove
859	8830 W Sunset Blvd	90069	Sunset Strip	Remove

The following 6 fractional addresses ('1/2' and '1/4' format) are valid in Los Angeles but warrant attention. Two have formatting anomalies that require correction.

Table 3 — Fractional Address Formats ('1/2' and '1/4')

Stop #	Address As Listed	Format	Status	Action Required
55	1514 1/2 Sunset Plaza Dr	Standard 1/2	Valid LA ADU format	Note for homeowner profile
371	3548 1/2 Multiview Dr	Standard 1/2	Valid LA ADU format	Note for homeowner profile
594	1801 1/2 S Sycamore Ave	Standard 1/2	Valid LA ADU format	Note for homeowner profile
634	1242 1/2 Gordon St	Standard 1/2	Valid LA ADU format	Note for homeowner profile
665	624 Wilcox 1/2 Ave	Malformed	Parsing error — fraction appended to street name	Correct to '624 1/2 Wilcox Ave'
706	617 1/4 N Sycamore Ave	Unusual 1/4	Atypical even for LA	Manual address verification required

1C. Sale Recency Analysis

Table 4 — Sale Recency Distribution Across 867 Stops

Category	Label Range	Approx. Count	% of Route	Notes
Just Sold	0m	1	0.1%	Stop #459 — HIGHEST PRIORITY IN ENTIRE ROUTE
Fresh	1d–30d	~142	~16.4%	Includes 5d, 6d, 7d, 9d–30d labels
Recent	1m–3m	~218	~25.1%	Within default lookback window
Aging	4m–9m	~341	~39.3%	Beyond default 3-month lookback
Stale	10m–12m	~166	~19.1%	Beyond default 3-month lookback
TOTAL	—	868	100%	—

Stop #459 — 1207 Greenacre Ave, West Hollywood, CA 90046. Recency label '0m' — sold THIS MONTH. This is the single hottest lead in the entire 867-stop route. The homeowner just closed escrow and is in the peak window for solar, home services, insurance, and related outreach. Knock this door before any other stop.

Notable fresh leads (leq30 days) in the early stops of the route:

- Stop 5: 7933 Fareholm Dr — 15d
- Stop 6: 8080 Laurel View Dr — 5d
- Stop 7: 8171 Mannix Dr — 13d
- Stop 19: 8518 Appian Way — 11d
- Stop 24: 2127 Kress St — 12d
- Stop 28: 2310 Stanley Hills Dr — 21d
- Stop 29: 8050 Jovenita Canyon Dr — 6d
- Stop 31: 2180 Laurel Canyon Blvd — 13d
- Stop 32: 2154 Beech Knoll Rd — 6d
- Stop 40: 1700 Queens Ct — 20d
- Stop 66: 8411 Hollywood Blvd — 6d
- Stop 76: 651 N La Jolla Ave — 28d
- Stop 77: 8215 Clinton St — 15d

Fresh lead concentration by neighborhood (leq30 days):

Table 5 — Fresh Lead Concentration by Neighborhood

ZIP / Area	Approx. Fresh Stops (leq30d)
90046 — Hollywood Hills / Laurel Canyon	~28
90068 — Lake Hollywood / Cahuenga Pass	~22
90036 — Fairfax / Mid-City	~18
90048 — WeHo / Beverly Center	~16
90035 — Beverlywood / Pico-Robertson	~14

Critical Finding: Fresh leads are distributed roughly evenly throughout the route — approximately 27% in stops 1–200, 25% in stops 201–400, 23% in stops 401–600, and 26% in stops 601–867. They are NOT front-loaded. Reps may spend the first half of the day knocking stale 10–12 month leads before reaching fresh prospects. Recency-weighted reordering within each neighborhood cluster is strongly recommended.

1D. Data Anomalies

1. Extended lookback anomaly: 58.4% of stops (~507) show recency >3 months, inconsistent with the default 3-month lookback. See Section 3A for full root cause analysis.
2. ZIP code outliers at route periphery: 91604 (Studio City, ~5 miles north over Mulholland Drive), 90025 (West LA, ~6 miles west of origin), 90024 (Westwood, ~5 miles west) — these ZIPs are at the outer edge of the coverage area and may warrant separate sub-routes.
3. Stop #706: '617 1/4 N Sycamore Ave' — A '1/4' fractional address is highly unusual even for Los Angeles. Flag for manual address verification before knocking.
4. Stop #665: '624 Wilcox 1/2 Ave' — Likely a data parsing error. The correct format should be '624 1/2 Wilcox Ave.' The fraction has been appended to the street name rather than the house number.

1E. Summary Statistics

Table 6 — Route 1 Data Quality Summary

Metric	Count	Notes
Total stops in file	867	—
Confirmed duplicate stops	~52	7 repeated data blocks in file
Invalid addresses	2	Stop #490 (house# = 0), Stop #848 (route origin)
Commercial addresses flagged	8	Stops 45, 58, 59, 67, 68, 803, 858, 859
Just sold (0m)	1	Stop #459 — knock immediately

Fresh leads (leq30 days)	~142	16.4% of route
Recent leads (1–3 months)	~218	25.1% of route
Aging leads (4–9 months)	~341	39.3% of route — beyond default lookback
Stale leads (10–12 months)	~166	19.1% of route — beyond default lookback
'1/2' / '1/4' addresses	6	Valid but flag for homeowner profile review
Recommended removals	~62	Dupes + invalids + commercial
Clean stops after audit	~805	—

Section 2 — Route Optimization Analysis

2A. Geographic Coherence Assessment

The current route sequence is highly fragmented. The 2-opt TSP solver appears to have been run on the full 867-stop set without neighborhood pre-clustering, which significantly degrades TSP performance at this scale. Estimated major geographic jumps (greater than 2 miles between consecutive stops): approximately 35–45 instances. The 10 most severe confirmed teleportation jumps are documented below.

Table 7 — Top 10 Confirmed Geographic Teleportation Jumps

Jump #	From Stop	From Address	To Stop	To Address	Est. Distance	Direction
1	259	9053 Elevado St, WeHo 90069	260	609 N Oakhurst Dr, Beverly Hills 90210	~2.5 mi	West
2	268	717 N Alta Dr, Beverly Hills 90210	269	1143 Sierra Alta Way, LA 90069	~2.0 mi	East (backtrack)
3	283	8882 Lookout Mountain Ave, 90046	284	2030 Oakstone Way, 90046	~1.5 mi	Different canyon
4	340	Hollywood Hills area	343	Studio City (91604)	~4.0 mi	North over Mulholland

5	400	7349 Pacific View Dr, 90068	401	7324 Woodrow Wilson Dr, 90046	~0.8 mi	ZIP boundary cross
6	446	2003 La Brea Ter, 90046	447	7360 Hollywood Blvd, 90046	~1.8 mi	Large jump same ZIP
7	528	Beverly Hills area	529	90005 Koreatown	~5.0 mi	East (major backtrack)
8	769	2101 Thurman Ave, 90016 West Adams	770	5832 David Ave, Culver City 90034	~3.0 mi	West
9	786	1318 S Roxbury Dr, 90035	787	443 S Peck Dr, Beverly Hills 90212	~1.5 mi	West into BH
10	803	9674 Wilshire Blvd, BH 90212	804	1714 Warnall Ave, 90024 Westwood	~3.0 mi	West

2B. Street Sweep Pattern Evaluation

A proper street sweep visits all stops on a given street consecutively, in ascending or descending address number order, before moving to the next street. The following failures were identified in Route 1:

1. Kirkwood Dr — CRITICAL FAILURE (845-stop gap)

- Stop 10: 8215 Kirkwood Dr
- Stop 11: 8237 Kirkwood Dr
- Route abandons street — 23 stops pass on other streets
- Stop 35: 8461 Kirkwood Dr
- Route abandons street again — 820 stops pass
- Stop 856: 8329 Kirkwood Dr
- Stop 856 (8329) should have been visited between stops 10 (8215) and 11 (8237) in ascending order. This is the most severe sweep failure in the route.

2. Woodrow Wilson Dr — Backtracking Within Street

- Stop 401: 7324 Woodrow Wilson Dr
- Stop 402: 7319 Woodrow Wilson Dr — backtrack (lower number)
- Stop 403: 7321 Woodrow Wilson Dr — backtrack again

- Stop 408: 7175 Woodrow Wilson Dr — jump down
- Stop 409: 7166 Woodrow Wilson Dr — descending (OK)
- Stop 410: 7240 Woodrow Wilson Dr — jump back up (backtrack)
- The route oscillates on this street rather than sweeping cleanly in one direction.

3. Pacific View Dr — 16-Stop Gap

- Stop 400: 7349 Pacific View Dr
- 16 stops on other streets intervene
- Stop 416: 7308 Pacific View Dr
- Stop 419: 7028 Pacific View Dr
- Stop 420: 6894 Pacific View Dr
- All four stops should be consecutive. The street is visited, abandoned for 16 stops, then resumed.

4. Multiview Dr — Direction Reversal

- Stop 353: 3587 Multiview Dr
- Stop 354: 3589 Multiview Dr — ascending (OK)
- Stop 356: 3740 Multiview Dr — ascending (OK)
- 13 stops on other streets intervene
- Stop 369: 3575 Multiview Dr — jumps back below 3587
- Stop 370: 3564 Multiview Dr — descending
- Stop 371: 3548 1/2 Multiview Dr — descending
- The route ascends to 3740, then reverses direction and descends back through addresses it passed earlier. Major sweep reversal.

5. Appian Way — Minor Out-of-Order

- Stop 19: 8518 Appian Way
- Stop 20: 8547 Appian Way — ascending (OK)
- Stop 21: 8655 Appian Way — ascending (OK)
- Stop 22: 8600 Appian Way — should come BEFORE stop 21
- Stop 22 (8600) should precede stop 21 (8655) for proper ascending sweep. Minor inefficiency.

2C. Neighborhood Clustering Analysis

The route does NOT cluster neighborhoods efficiently. Key ZIP codes are fragmented across multiple non-contiguous clusters, forcing the rep to repeatedly cross the same geographic areas. Only ZIP 90004 (Hancock Park/Larchmont) is visited in a single contiguous cluster.

Table 8 — Neighborhood Clustering Assessment by ZIP Code

ZIP	Area	Est. Stops	Sequence Spread	Clustered?
90046	Hollywood Hills / Laurel Canyon	~95	Stops 1–39, 62–75, 283–284, 366–403, 446–480, 849–867	NO — 6 clusters
90068	Lake Hollywood / Cahuenga Pass	~85	Stops 351–445, 712–737	Partial — 2 clusters
90036	Fairfax / Mid-City	~75	Stops 465–515, 558–563, 675–712	Partial — 3 clusters
90035	Beverlywood / Pico-Robertson	~70	Stops 117–222, 779–826	NO — 2 clusters
90034	Palms / Culver City adj.	~65	Stops 130–166, 770–832	NO — 2 clusters
90019	Mid-City / West Adams	~60	Stops 167–184, 517–602	Partial — 2 clusters
90048	WeHo / Beverly Center	~55	Stops 76–77, 216–250, 835–846	NO — 3 clusters
90069	West Hollywood Hills	~50	Stops 40–71, 847–863	Partial — 2 clusters
90004	Hancock Park / Larchmont	~50	Stops 615–690	GOOD — 1 cluster
90038	Hollywood / East Hollywood	~45	Stops 628–673, 713–721	Partial — 2 clusters
90005	Koreatown / Mid-Wilshire	~30	Scattered	Poor
90020	Koreatown	~20	Scattered	Poor
91604	Studio City	~18	Stops 340–350	Good — isolated
90210	Beverly Hills North	~15	Stops 103–116, 230–235, 260–268, 787–802	NO — 4 clusters
90211	Beverly Hills South	~15	Scattered	Poor
90212	Beverly Hills South	~20	Scattered	Poor
90024	Westwood / Brentwood	~12	Stops 804–815	Partial
90025	West LA / Sawtelle	~8	Scattered	Poor

90064	Rancho Park / Cheviot Hills	~10	Scattered	Poor
-------	-----------------------------	-----	-----------	------

Optimal neighborhood visit sequence from 911 N Kings Rd (WeHo origin):

1. 90069 — West Hollywood Hills (immediate vicinity of origin)
2. 90048 — Beverly Center / WeHo South (adjacent south)
3. 90046 — Hollywood Hills / Laurel Canyon (north/northeast)
4. 90068 — Lake Hollywood / Cahuenga Pass (northeast)
5. 91604 — Studio City (over Mulholland — do all at once)
6. 90036 — Fairfax / Mid-City (return south)
7. 90004 — Hancock Park / Larchmont (east)
8. 90038 — Hollywood / East Hollywood (northeast of Hancock Park)
9. 90028 — Hollywood Blvd corridor (adjacent)
10. 90005 / 90020 — Koreatown / Mid-Wilshire (southeast)
11. 90019 — Mid-City / West Adams (south)
12. 90016 — West Adams (further south)
13. 90035 — Beverlywood / Pico-Robertson (southwest)
14. 90034 — Palms / Culver City adjacent (west)
15. 90064 — Rancho Park / Cheviot Hills (west)
16. 90025 — West LA / Sawtelle (far west)
17. 90024 — Westwood (far west)
18. 90210 / 90211 / 90212 — Beverly Hills (do all BH together — southwest of origin)

2D. Recency-Weighted Optimization

Fresh leads (leq30 days) are distributed approximately evenly throughout the route — approximately 27% in stops 1–200, 25% in stops 201–400, 23% in stops 401–600, and 26% in stops 601–867. They are not front-loaded. The following recency-weighted reordering strategy is recommended:

- Stop #459 (0m — 1207 Greenacre Ave, West Hollywood 90046) should be the absolute first stop of the day, regardless of geographic sequence.
- All leq15d stops should be prioritized within their neighborhood cluster — visit them first before moving to older leads in the same area.
- Stops >6 months old should be placed at the end of each sub-route or flagged for a separate 'stale sweep' session.
- Apply recency-weighted sort within each sub-route after TSP optimization: 0m first, then leq15d, then leq30d, then 1–3m, then 4m+.

2E. Specific Optimization Recommendations

Stops to Remove Immediately:

1. Stop #490 — 0 N Alta Vista Blvd, Los Angeles, CA 90036. Invalid record (house number = 0). Delete from route.

2. Stop #848 — 911 N Kings Rd, West Hollywood, CA 90069. Route origin address. Not a sales prospect. Delete from route.
3. All confirmed duplicate stops from the 7 repeated data blocks (~52 stops total): second occurrences of Stops 137–146, 172–174, 377–386, 659–668, 694–703, and 797–803 blocks.
4. Stops 45, 58, 59, 67, 68, 803, 858, 859 — commercial addresses. Flag for manual review before removal; Hollywood Blvd stops may contain residential units above commercial ground floors.

Estimated efficiency gain: 35–45% reduction in total walking/driving distance if re-optimized with neighborhood pre-clustering and a fresh 2-opt TSP run per sub-route. At 867 stops, this mega-route represents 8–34 standard FirstKnock routes worth of data and should be split into 8 sub-routes of approximately 100 stops each.

Section 3 — Pipeline Integrity Check

3A. Extended Lookback Anomaly

Approximately 507 stops (58.4%) show recency labels of 4 months or older, which is inconsistent with the default 3-month lookback window in FirstKnock's ZipCodeExplorer. Three possible root causes are identified:

1. ZipCodeExplorer 'Sold within' filter set to 1 year — The most likely cause. If the operator set the lookback to 12 months instead of 3 months when generating this route, all properties sold within the past year would be included. This is the first hypothesis to test.
2. CSV imports via CsvUploader bypassing the pipeline — Records imported via CsvUploader with `data_source: 'csv_import'` bypass the standard BatchData/RentCast pipeline and may carry arbitrary recency values from the source CSV.
3. Legacy data predating current pipeline — Older records in MasterProperty that predate the current pipeline configuration may have been included in the route generation query.

Recommended action: Query MasterProperty WHERE `recency > 90 days` AND `property_hash IN (route stop list)` GROUP BY `data_source` to determine the split between 'rentcast', 'batchdata_verified', and 'csv_import' records. This will identify the root cause and inform whether a re-pull or a filter correction is needed.

3B. BatchData Confidence Tiers

With 867 stops processed across multiple chunks, the 100-call BatchData cap per processing chunk means that AMBIGUOUS properties in each chunk were classified by heuristic score only (`sale_confidence: 'medium'`). These stops have lower address verification confidence than 'batchdata_verified' records (`sale_confidence: 'verified'`). At route scale, a meaningful percentage of stops may carry medium confidence only.

Recommended action: Flag medium-confidence stops in RepHome.jsx with a visual indicator (e.g.,

amber badge or '?' icon) so reps are aware when approaching a stop with lower verification confidence. This prevents wasted knocks on misclassified properties and helps reps calibrate their pitch expectations.

3C. Commercial Address Flags

Table 9 — Commercial Address Flags with Disposition Recommendations

Stop #	Address	ZIP	Commercial Corridor	Recommendation
45	8558 Hollywood Blvd	90069	Hollywood Blvd	Manual verify — may be residential above commercial
58	8804 Hollywood Blvd	90069	Hollywood Blvd	Manual verify — may be residential above commercial
59	8800 Hollywood Blvd	90069	Hollywood Blvd	Manual verify — may be residential above commercial
67	8383 Hollywood Blvd	90069	Hollywood Blvd	Manual verify — may be residential above commercial
68	8378 Hollywood Blvd	90069	Hollywood Blvd	Manual verify — may be residential above commercial
803	9674 Wilshire Blvd	90212	Wilshire Blvd (BH commercial)	Remove
858	8800 W Sunset Blvd	90069	Sunset Strip	Remove
859	8830 W Sunset Blvd	90069	Sunset Strip	Remove

Note: Hollywood Blvd between La Brea and Fairfax (90069) contains a mix of residential and commercial properties. Stops 45, 58, 59, 67, and 68 should be manually verified before removal — some may be residential units above commercial ground floors and represent valid prospects.

3D. Special Address Formats

Table 10 — Fractional Address Format Review

Stop #	Address As	Format	Status	Action Required
55	1514 1/2 Sunset Plaza Dr	Standard 1/2	Valid LA ADU format	Note for homeowner profile
371	3548 1/2 Multiview Dr	Standard 1/2	Valid LA ADU format	Note for homeowner profile
594	1801 1/2 S Sycamore Ave	Standard 1/2	Valid LA ADU format	Note for homeowner profile
634	1242 1/2 Gordon St	Standard 1/2	Valid LA ADU format	Note for homeowner profile
665	624 Wilcox 1/2 Ave	Malformed	Parsing error — fraction appended to street name	Correct to '624 1/2 Wilcox Ave'
706	617 1/4 N Sycamore Ave	Unusual 1/4	Atypical even for LA	Manual address verification required

Section 4 — Action Plan

Immediate Actions (Before Next Knock Session)

1. Remove Stop #490 — 0 N Alta Vista Blvd, Los Angeles, CA 90036. Invalid record (house number = 0). Delete from route immediately.
2. Remove Stop #848 — 911 N Kings Rd, West Hollywood, CA 90069. This is the route origin address. It is not a sales prospect. Delete from route immediately.
3. Remove confirmed duplicate stops — Delete the second occurrence of each repeated data block: Stops 137–146 (second occurrence), 172–174 (second occurrence), 377–386 (second occurrence), 659–668 (second occurrence), 694–703 (second occurrence), 797–803 (second occurrence). Total: ~52 stops removed.
4. Flag for manual review before knocking: Stops 45, 58, 59, 67, 68 (Hollywood Blvd commercial corridor), Stop 803 (9674 Wilshire Blvd, Beverly Hills commercial), Stops 858 and 859 (Sunset Strip commercial). Verify whether these are residential units before including in route.
5. KNOCK STOP #459 TODAY — 1207 Greenacre Ave, West Hollywood, CA 90046. Recency label '0m' — sold this month. This is the single highest-priority door in the entire 867-stop route. The homeowner just closed escrow and is in the peak window for solar, home services, insurance, and related outreach. Do not wait.
6. Correct address parsing errors: Update Stop #665 from '624 Wilcox 1/2 Ave' to '624 1/2 Wilcox Ave.' Flag Stop #706 (617 1/4 N Sycamore Ave) for manual address verification

before knocking.

Before Re-Running This Territory

1. Re-pull with explicit 3-month lookback — Set 'Sold within' to 3 months in ZipCodeExplorer before generating new routes for this territory. This will eliminate the ~507 stale stops (4m+) that are currently diluting route quality.
2. Split into 8 sub-routes by neighborhood cluster — See the sub-route split table below.
3. Re-run routeOptimizer.js on each sub-route independently — Running the 2-opt TSP solver on 100-stop sub-routes rather than the full 867-stop mega-route will produce dramatically better street-sweep sequences. TSP performance degrades significantly at scale without pre-clustering.
4. Apply recency-weighted sort within each sub-route — After TSP optimization, apply a secondary sort within each neighborhood cluster: 0m stops first, then leq15d, then leq30d, then 1–3m, then 4m+.

Table 11 — Recommended 8 Sub-Route Split by Neighborhood Cluster

Sub-Route	ZIPs	Est. Stops	Area
A	90069 + 90048	~105	WeHo Hills + Beverly Center
B	90046	~95	Hollywood Hills / Laurel Canyon
C	90068 + 91604	~103	Lake Hollywood + Studio City
D	90036 + 90004	~125	Fairfax + Hancock Park
E	90038 + 90028	~55	Hollywood
F	90005 + 90020 + 90019 + 90016	~118	Mid-City / Koreatown / West Adams
G	90035 + 90034	~135	Beverlywood / Palms
H	90210 + 90211 + 90212 + 90024 + 90025 + 90064	~80	Beverly Hills + West LA

Data Pipeline Recommendations

1. Investigate the extended lookback anomaly — Query MasterProperty WHERE recency > 90 days AND property_hash IN (route stop list) GROUP BY data_source. Determine what percentage of stale stops are 'csv_import' vs. 'rentcast' vs. 'batchdata_verified'. This will

- identify whether the issue is a filter setting or a data import problem.
2. Add route-level data freshness score — Add a `freshness_score` field (0.0–1.0, weighted by recency distribution) to the `SavedRoute.metrics` JSON field. Display this score in the route management UI so operators can assess route quality at a glance before assigning to reps.
 3. Add stale lead visual indicator in `RepHome.jsx` — Stops older than 6 months should display a visual indicator (gray badge, reduced opacity, or 'STALE' label) so reps can make informed decisions about whether to knock. This is especially important for the ~166 stops (19.1%) in the 10–12 month range.
 4. Add commercial address filter to `routeOptimizer.js` — Implement a filter that flags addresses on known commercial corridors: Wilshire Blvd (Beverly Hills section), Sunset Blvd (Sunset Strip section), Hollywood Blvd (commercial sections). Cross-reference against a commercial corridor bounding box or street segment list.
 5. Add deduplication check at route generation time — The `MasterProperty` table deduplicates by `address_hash`, but the route generation layer should also deduplicate `property_hashes` in `SavedRoute` before saving. The current duplication pattern (entire data blocks repeated) suggests the 'Combine All' function in `ZipCodeExplorer` is not deduplicating before merging route segments.
 6. Add maximum route size warning — Implement a warning in `ZipCodeExplorer` when 'Combine All' would produce a route exceeding 200 stops. Display: 'This route contains [N] stops. FirstKnock standard routes are 25–100 stops. Consider splitting into [N/100] sub-routes for optimal performance.' This prevents future mega-route generation.

Appendix — Stop #459 Priority Alert

Stop #459: 1207 Greenacre Ave, West Hollywood, CA 90046 — Recency: 0m — SOLD THIS MONTH. Knock this door before any other stop in the route.

Address	1207 Greenacre Ave, West Hollywood, CA 90046
Recency Label	0m — Sold this calendar month
Stop Number	#459 of 867

ZIP Code	90046 — Fairfax / Hollywood Hills area of West Hollywood
Distance from Origin	~1.3 miles from 911 N Kings Rd (route origin)
Recommended Sub-Route	Sub-Route B (90046 — Hollywood Hills / Laurel Canyon)

This is the single highest-priority door in the entire 867-stop Route 1. The recency label '0m' indicates the property sold within the current calendar month. The homeowner has just closed escrow and is in the peak window for outreach — they are actively moving in, making purchasing decisions, and highly receptive to home services, solar, insurance, security, and related products. The conversion probability for a 0m lead is significantly higher than any other stop in this route. This stop should be knocked before any other stop in the route, regardless of geographic sequence. Do not wait for a scheduled route day — knock this door at the earliest opportunity. The window for peak receptivity narrows rapidly after the first 30–60 days post-close.

Document prepared for FirstKnock Sales OS operational review. All stop counts and recency figures are based on analysis of the 867-stop Route 1 file. Duplicate counts are approximate pending automated deduplication. Commercial address flags are based on known corridor geography and should be manually verified before removal. Contributors: Nick (Lead Developer), Danny (@daannyyrod).